

Terra Ops — Full Build Spec & Outstanding Backlog

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Purpose of this document: a complete, self-contained brief on what Terra Ops is, what already exists, and every single thing still to be built. Written so it can be handed to another AI or developer cold, with no prior conversation, and they can add detail or complexity to any section without breaking what's already there.

1. Executive summary

Terra Ops is one platform replacing **ServiceM8 + ActiveCampaign + the phone system** for a flooring business. Website feeds leads in, Xero takes the accounting, Stripe takes payments, an AI agent writes in the owner's voice, and installers work from a restricted mobile app that never shows them customer pricing.

Roughly 45% of the intended platform is built and working. The core job system, dispatch, tiering, availability, geo-tracking and the job-file media system are live. Still to build: a forms engine, financial integrations, CRM and email campaigns, SMS/voice with an AI agent, materials ledger with barcode scanning, and website lead capture.

Three hard constraints that must never be broken:

- 1. Terra Flooring, Floors by You and JSL Energy are 100% separate businesses.** Never combine, link, cross-reference or merge them. This spec is Terra Flooring only. Do not propose consolidation — the answer is permanently no.
 - 2. Installers must never see customer pricing, margins, quotes, invoices, supplier costs, or other installers' tasks and rates.** Enforced server-side, not in the UI.
 - 3. Installers on a lower rating tier are never told a job went to a higher tier.** They only ever see *"Sorry, this job was already accepted."*
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2. Tech stack (fixed — do not substitute)

Monorepo, Bun workspaces + Turborepo.

Layer	Technology
API	oRPC procedures on Hono, served at <code>/api/rpc/*</code>
Database	Drizzle ORM over Turso (SQLite)
Web frontend	React 19 + Wouter + Tailwind CSS 4, bundled by Vite 7
Mobile	Expo + React Native + expo-router (SDK 54)
Desktop	Electron shell loading the web app
Storage	Tigris (S3-compatible), private objects with signed GET/PUT
Auth	Better Auth — email/password + Google
Runtime	Bun

One codebase, one database. `packages/web` is the office system and the API. `packages/mobile` is the installer app ("Terra Crew"). They share one database. Mobile only ever calls `orpc.field.*` and `orpc.upload.presign`.

Fixed ports: web 4200, mobile 4300, desktop 4400.

3. Design direction

Approved direction: **dark and high-end — charcoal sidebar and headers, gold accents, like a premium dashboard**. Explicit instruction: *no cartoon-looking figures or avatars anywhere* (a criticism of ServiceM8's dispatch board). Crew map pins are coloured circles with initials, never illustrated people.

Token	Value
Font	Manrope
--gold	#bc9558
--gold-deep	#906f3c
--gold-pale	#c4b99f
--gold-wash	#f6f1e8
--charcoal	#161513
--background	#f4f2ef
--primary	#bc9558 on #171614

The overall feel should read "a bit like ServiceM8 but not exactly" — same density and speed of a trade dispatch tool, but premium rather than utilitarian.

Open decision: the mobile app is still terracotta (#C05A2B) and clashes with the gold/charcoal web system. Whether to retheme mobile to match is unresolved.

4. Data model — the locked-in decisions

These were decided deliberately and should be treated as fixed foundations. Add to them; don't rewrite them.

Contacts and companies

- **Contacts are people. They exist once, forever.** No duplicate contact per job.
- **Companies are an optional wrapper** over contacts (`company_contacts` join). A person can belong to a company or stand alone.
- Billing is set per job: `jobs.billToType` = "company" | "contact" .
- Jobs have **many contacts with roles** (`job_contacts`), each with their own flags: `receivesSms` , `receivesEmail` , `canApproveQuote` , `onSiteContact` . So a builder, a tenant and a strata manager can all sit on one job with different permissions.

Jobs and tasks

- **Tasks are the dispatch unit, not jobs.** One job produces many dispatches. Each task carries its own skill, crew size, assigned installer and pay. A single job might be "screed Monday, lay Tuesday, cove

Thursday" — three tasks, potentially three different installers.

- Job statuses, skills and rates are **editable data in Settings**, never hardcoded enums.

Installers and skills

- **Skills are ticks on the installer card** (`installer_skills`), each with a per-skill rate and a `canLead` flag.
- Tasks are only ever offered to installers ticked for that skill.
- Crew capacity per installer: `solo` / `own_offsider` / `needs_partner` .
- A furniture flag on a task **forces crew size 2**.
- Installers always see the customer's name and phone — except in tender mode.
- One pricing tier for customers.

Dispatch and tiering

- Dispatch is **direct** (to one installer) or **broadcast** (to everyone with the skill).
- Broadcast: **first accept wins**, all other offers auto-close to `"filled"` .
- Offers **expire after 2 hours**, then escalate to the owner.
- `INSTANT_ACCEPT_MIN_STARS = 4` . Installers rated 3 stars and under get a `provisional` accept held for 2 hours, letting a higher-rated installer take it first.
- Losing installers **only ever** see *"Sorry, this job was already accepted."*

Auth and roles

- Roles: `admin` | `installer` .
- The first user ever to sign in is auto-promoted to admin; everyone after defaults to installer and must be linked to an installer record by an admin.
- Disabled accounts are refused at the middleware.
- Self-protection: an admin can't demote or disable their own login.

5. What is already built and working

Phase 1 — core job system ☐ DELIVERED

Contacts, companies, sites, jobs, tasks, installers, skills, quotes, quote builder, a drag-and-drop schedule board with a dispatch panel, offers, a dashboard, settings, logins/team management, and database backups.

Installer app: login, Today / Coming up / Offers (with badge) / Me tabs, task detail with start/stop, checklist ticking, field notes, and own-pay display.

Phase 2 — in progress

Feature	Status
Rating tiers + 2-hour provisional hold + day-lock	☐ Done
Blackout dates / installer unavailability	☐ Done
On-shift geo location + live crew map	☐ Done
Job file — 7 media buckets + gates	☐ Done, just shipped
Plan-on-job-card	⚠ Schema only — superseded by the <code>plan</code> bucket, needs reconciling
Tender mode	⚠ Flag only, no enforcement
Materials / barcode / installer account	⚠ Schema only
Date-window fields on the office task form	☐ API accepts them, no admin UI

Geo location (done — the rules matter)

Tracking runs **only** between "Start job" and "Mark complete". Each installer opts in themselves on the Me tab. Switching it off deletes their entire history. History auto-deletes after 14 days. A pin goes stale after 45 minutes. Nobody appears on the office map until an installer has opted in *and* started a job.

The job file (done — the model to extend)

Replaces ServiceM8's diary feed, which was described as *"messy and all over the place"*. **Seven fixed named buckets**, not a chronological feed:

Bucket	Purpose
plan	Plans — office uploads, crew read-only
access	Site access — parking, lift, gate code, stairs
area	Areas we're doing — room by room
damage	Existing damage — before anything starts
found	What we found — subfloor, moisture, nasties. Variation trigger.
completion	Finished work
defect	Defects and callbacks

Rules in force:

- Photos, video **up to 60 seconds**, and PDFs in any bucket.
- **Crew can add, never delete.** The office archives instead of deleting, so the record survives a dispute.
- **Cannot start a job** until damage photos are added or "nothing found" is ticked — that tick is stamped with who and when.
- **Cannot mark complete** until completion photos meet the per-skill minimum (default 4, editable in Settings).
- Everything auto-stamped: uploader, timestamp, GPS, task, area. Nobody names a file.
- Anything dropped in `found` writes a `site_finding` activity entry against the job — the hook the variation form plugs into.
- Files go phone/browser **straight to Tigris**, never through the API. Links are signed on read; nothing sits on a public URL.

Note: signature-on-completion was deliberately **not** made a blocker.

6. THE BACKLOG — everything still to build

This is the part to expand. Ordered by the sequence agreed with the owner.

6.1 Forms engine + variation form ☐ NEXT

Stated need: *"we need forms like variation one."*

Proposed tables:

- `formTemplates` — name, kind, active

- `formFields` — `templateId`, `label`, `type` (`text` / `number` / `select` / `checkbox` / `photo` / `signature`), `required`, `options JSON`, `sortOrder`
- `formSubmissions` — `templateId`, `jobId`, `taskId`, `installerId`, `submittedAt`, `status`
- `formAnswers` — `submissionId`, `fieldId`, `label`, `value`

First template — **Variation**: what changed, why, extra m² or hours, photos auto-linked to the `found` bucket. It lands in the office as a **pending variation the admin prices**. Crew describe the work; they never see or enter a customer price.

Hooks into the existing `site_finding` activity entry. The form builder lives in Settings, following the existing "editable data, never hardcoded enums" pattern.

Obvious follow-on templates once the engine exists: pre-start site check, moisture/subfloor report, defect report, safety/SWMS, toolbox talk, vehicle check.

6.2 Tender mode lockdown

A tendered task shows the installer **plan, their pay, and photos only** — no address, no customer name, no phone. It unlocks the moment they win the job. Must be **enforced server-side** in the field routes, not just hidden in the UI. The newly built `jobFile` procedure currently returns everything and must be made tender-aware.

6.3 Office task form gaps

The API already accepts `scheduledFrom`, `scheduledTo` and `tenderMode`, but no admin screen sets them. This is an outstanding promise.

6.4 Reconcile plans

`jobs.planUrl` / `planName` / `planMime` and the per-task overrides predate the `plan` media bucket and now overlap. Pick one source of truth and migrate.

6.5 Offline capture queue

Sites frequently have no signal. Photos and form submissions need to queue on the device and sync when back in range, with visible pending state. Already promised to the owner.

6.6 Customer pack (before/after)

One tap turns `damage` + `area` photos into "before" and `completion` into "after", producing a shareable pack for the customer. The API endpoint exists (`media.customerPack`); the UI does not.

6.7 Auto job cover photo

Each job picks a hero image automatically from its media so the job list reads visually.

6.8 Materials, barcode scanning and installer accounts

The owner scans a product barcode (Sika self-leveller, glue) and sets an installer price. The installer scans it in the app, sees the price, and either pays on the spot (Stripe, later) or **puts it on account**.

- **Invoiced weekly and paid separately — never deducted from installer pay.**
- Per-installer dollar credit limit; blocked at the limit.
- Running balance visible to both sides.
- `installer_account_entries`, `products.barcode` and `products.installerPrice` already exist in the schema.
- Needs `expo-camera` for the scanner.

6.9 Payments and accounting — Stripe + Xero

Explicitly deferred: *"build everything else now, wire Stripe in later."*

- Stripe for deposits, progress claims and final invoices.
- **One-way push into Xero** — Terra Ops is the system of record, Xero is the ledger. No two-way sync.
- The owner has said he wants **control over what the customer-facing invoice looks like**, rather than accepting a default Stripe-hosted page.

6.10 CRM + email campaigns (ActiveCampaign replacement)

Lists, segments, templates, sends, opens and clicks. Built on the existing contact model — contacts already exist once and forever, which is the right base for segmentation. Needs automation/sequence logic (trigger → wait → send → branch), not just broadcasts.

6.11 SMS, voice and the AI agent

- Per-contact `receivesSms` / `receivesEmail` flags already exist on `job_contacts` and should drive all sending.
- **An AI agent that writes in the owner's voice** — quotes, follow-ups, replies.
- Phone strategy is **parked**: whether to port the existing mobile number or take a new Twilio number. Note for planning: an alphanumeric sender ID like `TERRAFLOOR` is one-way only and requires ACMA Sender ID Register approval as of 1 July 2026. Recommendation on the table was a new number now, port later.
- Considered but undecided: a 1300 number plus a mobile for "test us and we'll come back with a quote".

6.12 Website lead capture

Landing pages feed straight into Terra Ops as leads, replacing the current arrangement where lead pages sit outside the system and get pushed into ServiceM8. Several landing pages already exist as separate projects and would become sources.

7. Cross-cutting things not yet designed

Areas where added complexity would genuinely help, and where nothing has been decided:

- **Notifications** — who gets told what, on which channel, and how alerts escalate when an offer expires or a variation lands.
 - **Reporting** — job profitability, installer performance, m² per day, defect rate by installer or product, lead source ROI.
 - **Ratings** — the tiering system consumes a star rating, but how ratings are actually earned and updated is not defined.
 - **Warranty and defect lifecycle** — a defect bucket exists, but there's no callback workflow, SLA or warranty period tracking.
 - **Scheduling intelligence** — travel time, geographic clustering, capacity forecasting, weather.
 - **Document generation** — quotes, variations, completion certificates as branded PDFs.
 - **Audit trail** — an activity log exists per job, but there's no global immutable audit view.
 - **Permissions granularity** — only two roles today. An office admin vs. an estimator vs. the owner may eventually need different access.
 - **Customer-facing portal** — customers currently have no login. Approving a quote or variation online is unbuilt.
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8. Housekeeping and outstanding items from the owner

- Private GitHub repo `terra-ops` and its URL — not yet created.
 - Real installer names and rates; the real job status list; a genuine quote/invoice sample to model documents on.
 - Decision on whether the gold retheme carries into the mobile app.
 - Phone strategy decision.
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9. Notes for whoever picks this up

- **Read requests literally and narrowly before widening scope.** Scope creep is not welcome.
- When the owner answers a question with "No Preference" or "Yes", that means *you decide* — make the call and move on rather than asking again.
- Ship each piece as it's finished rather than batching. Lint and build after every change.
- Skills, statuses and rates are configuration, never code constants. Follow that pattern for anything new.
- Nothing that reveals price, margin or another installer's business ever crosses into the mobile app, and the enforcement lives on the server.

10. Admin-only estimating, price book & quote automation

This section adds a fast on-site estimating system to the Terra Ops admin job card. It replaces the slow spreadsheet workflow while preserving a reliable, auditable calculation engine. It is part of the core backlog and must be built without weakening the existing server-side installer restrictions.

Locked decision - AI assists retrieval, never arithmetic

The AI bar may understand natural-language searches and synonyms, but every quantity, rate, pack calculation, supplier fee, trim cut, GST amount and margin is calculated deterministically from stored structured data. The AI must never invent a product, price or measurement.

10.1 Outcomes

- An admin can build a complete estimate on site from one responsive section of the job card.
- Common measurements are entered directly: m2, lineal metres, each, hours, bags, rolls, packs, boxes and custom units.
- Commercial vinyl captures floor area, coving, internal and external corners, weld length, skim/levelling, stairs and other preparation.
- Customer selling prices, installer rates, supplier costs and job-specific overrides remain separate.
- Products and rooms stay attached to the correct work package so descriptions never mix carpet areas with vinyl or timber areas.
- The completed estimate generates a branded quote draft that remains fully editable before sending.

10.2 Non-negotiable security

- The estimator, supplier costs, selling prices, margins, quote editor and price book are admin-only and enforced server-side.
- Installers receive only their own assigned pay line items and their own total pay.
- No installer route or API response may include customer selling prices, margins, supplier costs or another installer's rates.
- All price and margin overrides record admin, time, previous value, new value and reason.

10.3 Core pricing model

Layer	Purpose	Change behaviour
Master suggested sell price	Default customer charge per item	Affects new jobs created after the change only
Highest eligible installer rate	Protected labour cost used while quoting	Calculated per estimate item from eligible installers
Installer-specific rate	Actual planned subcontract labour cost	Snapshots when installer is assigned/accepted
Supplier/material cost	Expected product and delivery cost	Snapshots into the job estimate
Job override	Admin-only exception for one job	Never writes back to the master price book

Snapshot rule

When a new job is created it receives the current price-book version. Later master changes apply to newly created jobs only. Existing jobs and quotes never change automatically. An admin may optionally preview and apply newer rates to a draft job, with a line-by-line difference report and audit record.

10.4 One-screen estimating workflow

Location: **Admin job card > Estimate & Quote**. The page is designed for a tablet or phone on site and desktop in the office. It autosaves as a draft.

Step	Admin action	System response
1	Search or browse a trade group	Returns exact price-book items, products and synonyms
2	Create a work package	Keeps product, installation, preparation and areas together
3	Enter quantities	Calculates labour, material usage, waste, packs and supplier fees
4	Assign areas beneath that product	Builds a product-specific scope sentence
5	Review cost and margin	Uses highest eligible installer rate until one is selected
6	Select/assign installer	Recalculates actual planned margin using that installer's rates
7	Generate quote	Creates editable line items and scope wording without losing cost links

Recommended screen layout

- Top: AI-assisted search bar with a manual group filter and recent/favourite items.
- Left/upper panel: alphabetically ordered trade groups. Items inside a group use admin-defined manual order, not alphabetical order.
- Centre: work-package cards containing product, areas, measurements, labour and preparation.
- Right/lower sticky summary: customer subtotal, materials, supplier fees, protected labour, gross profit, gross margin and warnings.
- Actions: duplicate package, reorder, split, archive, save draft, preview quote and generate quote.

Quick-entry behaviour

- Each item has a quantity box beside it. Example: Carpet install - 34 LM; Weld - 18 LM; Internal corner - 7 each.
- The unit is displayed beside the quantity and comes from the price-book item. Admin can override the unit only on a custom line.
- Keyboard-first entry on desktop; large tap targets, numeric keyboard and minimal modal windows on mobile/tablet.
- Autosave each edit. Show Saved, Saving and Offline/Pending states clearly.
- Allow optional notes, measurement source, area and attachment against every estimate line.

AI-assisted search

- Understands terms such as 'R12 vinyl', 'commercial safety vinyl', 'skim', 'smoothing compound', 'cove' and supplier/product aliases.
- Example query: '2 mm R12 Armstrong' filters by thickness, slip rating, brand and product category.
- Results display product, supplier, SKU, category, rating, dimensions/pack size, current cost and last-updated date.
- If nothing matches, say 'No matching item in the price book'. Never manufacture a result.

10.5 Price-book structure

The price book is editable configuration, not hardcoded data. It needs bulk import/export, active/inactive status, effective dates and an audit trail.

Entity	Required fields / behaviour
price_book_groups	name, description, groupSortOrder, active. Groups display alphabetically; sort order is retained for explicit future override.
price_book_items	groupId, name, aliases, SKU, unitType, itemType, suggestedSellRate, defaultBudgetCost, taxable, quoteTemplate, manualSortOrder, active
price_book_versions	version, effectiveAt, createdBy, reason. A job stores the version current at job creation.
installer_item_rates	installerId, priceBookItemId, rate, effectiveFrom, effectiveTo, active. Never exposed to another installer.
suppliers	name, account reference, standard dispatch rule, contact details, active
products	supplierId, brand, productName, SKU, category, attributes JSON, purchaseUnit, packSize, coverage, cost, lastUpdatedAt, active
job_work_packages	jobId, name, productId, workType, sortOrder, notes. Parent container that prevents cross-product area errors.
job_estimate_lines	packageId, item/product reference, description, quantity, unit, sell-rate snapshot, cost snapshots, override flags and totals
job_package_areas	packageId, areaType, label, count, notes. Areas belong to exactly one work package unless deliberately copied.
supplier_fee_rules	supplierId, rule type, threshold, fee, priority, effective dates and tax treatment
margin_approvals	jobId, triggeredByLine, marginBefore, marginAfter, approvedBy, approvedAt, mandatory reason

Units and item types

- Supported units: m2, LM, each, hour, day, bag, roll, length, pack, box, litre and custom.
- Item types: customer service, installer labour, material/product, supplier fee, allowance and custom.
- All monetary costs are stored consistently ex-GST for margin calculation; the quote renderer applies the configured customer GST display.
- A price may be left blank and completed on the job. Blank is not treated as zero.

Master versus job behaviour

- Editing a suggested price on a job changes only that job.
- Editing the master suggested price changes new jobs created from that point onward.
- Accepted quotes and dispatched task pay are immutable except through a named variation or audited admin adjustment.
- Deleting an item with history means archiving it, never removing the historical record.

10.6 Installer rates, eligibility & margin protection

Installer pricing is not a duplicate customer price list. It is a private cost schedule connected to the installer's skills and the price-book items they are permitted to perform.

Skills and visible rate groups

- An installer has skill toggles such as Carpet, Sheet Vinyl, Vinyl Plank, Hard Floor, Timber, Preparation, Stairs and Commercial Coving.
- Only applicable rate groups and items appear when editing that installer. Admin can add or remove individual item eligibility.
- Preparation can be linked as a companion group for carpet, vinyl and hard-floor installers, while still allowing individual prep skills to be disabled.
- Task eligibility continues to use installer_skills and canLead. Rate visibility does not grant a skill.

Quote-time protected labour

Owner decision - highest eligible rate

Before an installer is selected, the estimate uses the highest active rate among installers eligible for each labour line. This protects the quote from being profitable only when the cheapest installer is available. If no eligible installer has a rate, use the item's default budget cost and show a blocking 'No eligible installer rate' warning before the quote is sent.

Selecting an installer

- The admin may view eligible installers with availability, skill match, rating tier, distance/travel estimate, relevant rate and forecast job margin.
- The system may suggest alternatives, but must not recommend an installer solely because they are cheapest.
- After selection, actual planned labour cost uses that installer's snapshotted rates and the margin recalculates instantly.
- A later change to the installer's master rates does not alter already accepted/dispatched task pay.

Low-margin approval

Owner decision - warn and approve

If the selected installer or any estimate edit drops the job below the configured minimum gross margin, show a persistent red warning. The job may proceed only after an admin approves it and enters a reason. Do not automatically change the customer's price and do not block the installer from assignment.

Installer view

Owner decision - own line items and total pay

The installer sees their own pay line items, quantities, units, rates and calculated total for the assigned task, plus any approved pay variation. They never receive customer prices, profit/margin data, supplier costs or other installers' rates.

Margin calculation

- Gross profit = customer revenue ex-GST minus material costs, supplier fees, installer/subcontract labour and other direct job costs.
- Gross margin % = gross profit divided by customer revenue ex-GST.
- Show both protected margin (highest eligible installer) and selected-installer margin when an installer has been chosen.

10.7 Product catalogue & supplier rules

The material selector and AI bar search one structured product catalogue. Product records can carry different attributes by category without forcing every product into flooring-specific fields.

Product field	Examples / rule
Identity	Supplier, brand, product name, supplier SKU, Terra alias
Classification	Carpet, carpet tile, sheet vinyl, safety vinyl, LVT, timber, adhesive, leveller, trim, moisture barrier
Technical attributes	Thickness, width, roll length, slip rating, colour, coverage, density, pack quantity, stock length
Purchase data	Cost, purchase unit, minimum order, pack/box multiple, waste factor, last verified date
Quote data	Suggested sell price, sell unit, customer-facing name and description template

Supplier fee engine

- Combine all selected products from the same supplier before evaluating order thresholds.
- Apply a dispatch fee once per supplier order/job, not once per product line.
- Show the fee separately in cost calculations. Whether it appears separately or is absorbed into selling prices is configurable.
- Rules are effective-dated and may be flat, threshold-based, percentage, zone/postcode-based or manual.

Armstrong seed rule supplied by owner

If the Armstrong product subtotal is \$1,000 or less, apply a \$150 dispatch fee. If the subtotal is over \$1,000, apply a \$100 dispatch fee. The threshold, fee and tax basis remain editable configuration.

Material quantity calculations

- Coverage products calculate required packs/bags/rolls from measured quantity, depth where relevant, coverage and waste.
- Round purchases up to the supplier's purchase multiple while retaining the exact measured quantity for installation labour.
- Show calculated requirement, purchasable quantity, waste/overage, cost and manual override.
- A manually overridden purchase quantity is visibly flagged and requires a note when it is below the calculated requirement.

10.8 Areas belong to each product/work package

Every selected flooring product creates or joins a work package. The area selector appears directly beneath that product/package, never as one global room list for the entire job.

Standard area	Input behaviour
Lounge, Dining, Hallway, Passageway, Garage, Kitchen, Laundry	Tick box plus optional label/notes
Bedroom, WIR, Cupboard	Tick box plus quantity; optionally expand to Bedroom 1, Bedroom 2, etc.
Stairs	Tick box plus quantity and straight/winder/feature type
Custom area	Free-text name, quantity and notes

- Installation and preparation lines attach to the same work package or to explicitly selected areas.
- Moving an area between products requires confirmation if it already has installation or preparation lines.
- Copying an area to two products is allowed only as an explicit action, for example floor preparation plus finished flooring.

Quote wording example

Generated draft

Supply and install Armstrong R11 commercial sheet vinyl to the kitchen and hallway, including 100 mm covered skirting, internal/external corners and hot-welded seams.

- A second carpet product with bedrooms selected generates its own separate scope line. Its areas cannot be merged into the vinyl description accidentally.

10.9 Trim measurement & cut optimisation

Trim products store stock length and pack quantity. After selecting a trim type, the admin enters every required door/opening size beneath that exact trim product.

Input	Rule
Trim product	Exact SKU/profile/colour; only identical stock may share offcuts
Stock length	Supplier value, e.g. 3.3 m
Pack quantity	Lengths per box; purchasing rounds up to full boxes where required
Cuts	One row per doorway/opening: label, required length and quantity
Saw kerf	Configurable; initial suggested default 3 mm
Joining	Not allowed by default. May be enabled per trim/product if acceptable
Waste allowance	Configurable minimum plus actual optimisation waste

Calculation output

- Use a deterministic one-dimensional cutting-stock calculation to minimise full lengths while accounting for saw kerf.
- Return full lengths required, boxes required, total purchased length, used length, offcut/waste and total product cost.
- Produce a cut list per physical length: for example Length 1 - Door A 920 mm, Door B 820 mm, Door C 760 mm.
- Never combine offcuts across different SKU, profile, colour or job unless an admin deliberately adds known warehouse offcut stock later.
- Allow the admin to override calculated lengths/boxes, but preserve the original calculation and flag the override.

10.10 Quote generation and editing

- Generate quote sections in work-package order, with product, areas, installation, preparation, exclusions and optional notes.
- Customer wording uses quote templates and structured area data. It must not expose supplier cost names or installer pay.
- The admin can edit text, reorder lines, split/merge customer-facing sections, change quantities and change sell prices.
- Editing wording alone does not alter calculations. Editing a linked quantity or price recalculates totals and margin.
- The generated wording is a draft, not a locked template. Preserve the structured estimate behind the final edited text.
- Quote revisions are versioned. Accepted versions remain reproducible exactly as sent.

10.11 Acceptance tests

- Changing a master suggested price does not alter an existing job, quote or accepted task.
- Creating a new job after the change uses the new master price-book version.
- A job-specific price override never changes the master price book.
- Two products on one job retain independent area lists and generate correct separate quote descriptions.
- An R12 search returns only matching stored products and never invents a missing result.
- Armstrong supplier totals trigger the correct single dispatch fee across multiple Armstrong products.
- Trim optimisation respects 3.3 m stock lengths, kerf, pack multiples and SKU separation.
- Highest eligible installer rate is used before assignment; selected rate replaces it for actual planned margin.
- A below-minimum margin produces a red warning and cannot proceed without an admin approval reason.
- Installer API responses contain only that installer's own pay line items and total.

10.12 Initial installer-rate seed data

The following rows came from the attached owner note. Import them as **draft installer-cost items**, not customer selling prices. Admin must confirm each classification, rate and GST basis before activation. Blank prices remain incomplete and must not calculate as zero.

Carpet

Carpet item	Draft rate	Unit
3.6 m wide install	\$40	LM
4 m wide install	\$45	LM
3.6 m extra heavy	\$45	LM
4 m extra heavy	\$50	LM
Glue gripper	\$4	LM
Ramp MDF	\$15	LM
Capping	\$8	LM
Multi pinned/unpinned	\$8	LM
Stairs - straight waterfall	\$13	each
Stairs - winder waterfall	\$18	each
Stairs - straight bullnose	\$20	each
Stairs - winder bullnose	\$27	each
Feature step	\$50	each
Double-sided feature	\$100	each
Carpet tile install	\$10	m2
Carpet tile straight step incl. nosing, drill/plug/glue, max 1.2 m	\$60	each
Carpet tile winder step incl. nosing, drill/plug/glue, max 1.2 m	\$90	each

Vinyl

Vinyl item	Draft rate	Unit
Vinyl plank install	\$25	m2
Sheet vinyl install	\$28	m2
Cove - 100 mm	\$13	LM
Cove - 150 mm	\$18	LM
Cove - custom height	Blank	LM
Internal/external corner	\$15	each
Weld	\$8	LM
Straight stair with box-out	\$50	each
Winder stair with box-out	\$90	each
Square single bullnose with box-out	\$250	each
Square double bullnose with box-out	\$400	each

10.12 Initial installer-rate seed data - continued

Hard floor

Hard-floor item	Draft rate	Unit
Hybrid/laminate install	\$28	m2
Engineered timber click install	\$28	m2
Engineered T&G; floating with wood glue	\$35	m2
Engineered timber including glue	\$60	m2
Straight stair	\$50	each
Winder stair	\$100	each
Square single bullnose with box-out	\$250	each
Square double bullnose with box-out	\$400	each
Herringbone	\$100	m2 - confirm
Chevron	\$100	m2 - confirm
Scotia install	\$6	LM
Install new skirting	\$11	LM
Remove skirting	\$4	LM
Remove, de-nail and clean skirting	\$6	LM
Refit existing skirting	\$6	LM
Install trims	\$6	LM
Other	Blank	Custom

Preparation

Preparation item	Draft rate	Unit
Single skim	\$10	m2
Double skim	\$18	m2
Self-levelling at 4 mm, supply leveller	\$27	m2
Self-levelling, no leveller supplied	\$18	m2
Each additional mm	\$3	m2/mm
Light grind	\$5	m2
Medium grind	\$10	m2
Heavy grind	\$15	m2
Apply moisture barrier by trowel, product not supplied	\$7	m2
Ramp floor	Blank	each
Remove tiles, no disposal	\$30	m2 - confirm

10.13 Initial material-cost seed data

These are draft supplier/product cost records from the owner note. Empty prices require completion before use. Product names should be verified against current supplier invoices rather than silently corrected by AI.

Supplier	Product	Draft cost	Purchase unit / status
Intafloors	Mapei Adesilex G19	\$70.30	Confirm unit and GST basis
Intafloors	RLA Tuff Top	\$38.00	each; confirm pack size
Bayset	Uzin 182	Blank	Incomplete
Sika	Sika Level Top	\$19.50	Confirm bag size
Sika	Sikafloor-01 Primer (AU)	\$67.62	Confirm container size
Sika	SikaLevel SkimCoat C5001 / 4 x 4.5 kg	\$82.61	pack
Sika	Sikafloor-14 Prep N Prime 10 L	\$140.24	10 L

Outstanding owner input
Provide the complete customer selling-price list, installer items/rates, product catalogue, pack sizes/coverage, supplier fee rules, quote description templates, minimum margin setting and confirmation of whether each stored rate is ex-GST or inc-GST. The importer must preserve blanks and flag uncertain units for review.

10.14 Build sequence

Stage	Deliverable
1	Price-book groups/items, versions, units, import/export and audit trail
2	Installer item rates, skill eligibility and server-side visibility tests
3	Job work packages, quantities, per-product areas and snapshot logic
4	Product catalogue, supplier rules, material quantities and dispatch fees
5	Trim measurement, cut optimisation, pack rounding and cut-list output
6	Margin engine, highest-eligible-rate costing, red warning and approval audit
7	Quote description generation, full editing, versioning and branded PDF output
8	AI-assisted search over the completed structured catalogue

Do not build the AI search first. It depends on clean product and price-book data. Build and test deterministic pricing, snapshotting and permissions before adding natural-language retrieval.

11. Product surfaces: one database, three experiences

Terra Ops must present three deliberately different clients over the same API and database. This is not three separate systems. Each surface exposes only the work appropriate to its user and device. The permission boundary is enforced by the API, not by hiding buttons.

Locked product rule

Mobile is for doing and responding. Desktop is for configuring, analysing and building. Terra Crew is for completing assigned work. A new feature is not automatically added to all three surfaces.

11.1 Navigation

Surface	Primary navigation	Design goal
Terra Ops Desktop/Web	Dashboard, CRM, Jobs, Schedule, Inbox, Quotes, Marketing, Materials, Reports, Settings	Complete owner and office control
Terra Ops Admin Mobile	Home, Jobs, Schedule, Inbox, More, plus global search/action	Fast field quoting, communication and approvals
Terra Crew	Today, Coming Up, Offers, Me	Simple installer workflow with minimal decisions

11.2 Mobile vs desktop feature matrix

Capability	Desktop/Web	Admin Mobile	Terra Crew
Needs Attention	Full queue, filters, rule settings and bulk actions	Own/assigned alerts, quick actions and approvals	Own task blockers only
CRM and timeline	Full contact/company/job history and merge tools	Search, view, call, message and add notes	Assigned-task contacts and notes only
Leads and jobs	Create, edit, bulk manage and report	Create/search jobs, capture visits and update status	No lead management
Schedule and dispatch	Full board, capacity, travel and broadcast controls	Today/week view, assign, offer and contact	Own accepted work and offers
Estimating and quotes	Full build, review, templates, revisions and reporting	Measure, estimate, edit, approve and send on site	No customer quote access
Job execution	Review/configure forms, files, variations and completion	Capture, review and approve from the field	Start, photos, forms, variation and complete

11.2 Mobile vs desktop feature matrix - continued

Capability	Desktop/Web	Admin Mobile	Terra Crew
Inbox/email/calls/SMS	Full team inbox, routing, templates and rules	Compose, reply, search, link and create follow-up	Contact office; permitted job communication only
Marketing	Segments, campaigns, sequences, consent and reports	Summary/urgent approval only; no campaign builder	None
Price book/products	Configure, version, import/export and supplier rules	Search and use job snapshots; job-only overrides	Permitted material scan/account price only
Installers and pay	Profiles, skills, rates, eligibility and reporting	Assign and approve within permission profile	Own pay lines and own total only
Payments/accounting	Invoices, reconciliation, Xero/Stripe controls	Job payment state and authorised field actions	Own material account only
Reports/settings/audit	Full configuration, audit, exports and recovery	Critical alerts and named approvals only	None
AI search/action	Search allowed business data and draft actions	Fast field search and safe actions	Own assigned work only

11.3 Surface acceptance rules

- Every API procedure authorises the user, role, admin permission profile and record scope before returning data.
- Terra Ops Admin Mobile is a responsive admin experience; it is not the desktop sidebar compressed onto a phone.
- Marketing builders, bulk imports, master price configuration, integrations, permissions, detailed reporting and global audit remain desktop-only.
- Admin Mobile may show a critical summary and a 'Review on desktop' action when the full workflow is unsuitable for a phone.
- Terra Crew language uses one clear action per screen, large tap targets, plain instructions and no office jargon.
- Any future feature ticket must name its required surfaces. If unspecified, it goes to Desktop/Web only until product review.

12. Job lifecycle and Needs Attention engine

A job keeps its normal lifecycle stage while a separate rules engine calculates operational flags. A flag does not overwrite the job status. This prevents the schedule and reports being corrupted just to show that somebody needs to act.

12.1 Flag behaviour

- Red means overdue, blocked or likely to cost money. Amber means due soon. Blue/grey means legitimately waiting and has a next review date.
- Every flag has a rule, severity, owner, due time, reason, clear next action and link to the exact record.
- The system removes a flag automatically when its condition is resolved. Manual completion remains audited.
- Snooze requires a reason and a new date. Indefinite snooze is not allowed.
- Timers use configured Terra Flooring business hours and public holidays, not raw elapsed hours.

Trigger	Amber	Red / required action
New enquiry not contacted	After 2 working hours	End of business day - contact or record outcome
Job has no booking or next action	After 4 working hours	After 1 business day - book or set named next action
Site visit finished, no quote	Later that business day	Next business day - create/send quote or record pending reason
Quote drafted, not sent	Same-day reminder	After 1 business day - send, cancel or document blocker
Quote sent, no follow-up	Day 3	Day 7 - contact or set pending/lost outcome
Accepted quote/work order, no booking	Immediate	Immediate - remains red until booked
Upcoming work not ready	Configurable lead time	Deposit/materials/installer missing by cutoff
Work complete, evidence missing	At attempted completion	Block completion until required files/forms exist
Completed, invoice not sent	Same business day	Next business day - send or record accounts hold

12.2 Site visits and pending quotes

- Admin Mobile provides Start Site Visit and Finish Site Visit. Finishing the visit starts the quote timer.
- A pending quote requires: reason, who Terra is waiting on, assigned staff member, next follow-up date and notes.
- Standard pending reasons: customer deciding; insurance; builder/strata; product choice; revised scope; plans/measurements; finance; tender; customer requested later date; no response.
- Other red rules include overdue deposit, material not ordered, no installer near start date, expired/unaccepted offer, unpriced variation, unresolved defect and failed integration delivery.

12.3 Queue views

- Desktop Home shows My Attention, Team Attention and Unassigned, sortable by urgency, value, age, stage and owner.
- Admin Mobile Home shows the user's red items first, then today's visits/jobs and amber items.
- A daily summary may be sent, but critical red events also create an immediate in-app alert.

13. CRM, email and marketing automation

Contacts, conversations and jobs are separate records. Every email belongs to a mailbox and conversation; it may link to a contact; it may link to a job. A job link is never compulsory.

13.1 Unified customer timeline

- The contact timeline combines permitted calls, SMS, emails, website enquiries, site visits, quotes, jobs, payments/invoices, complaints, defects, marketing events and internal notes.
- Conversation categories: General enquiry, New work/quote request, Existing job, Accounts/payment, Complaint/defect, Supplier, Installer, Spam and Unclassified.
- Conversation statuses: Unassigned, Assigned, Waiting on us, Waiting on customer, Resolved and Archived.
- A recognised customer's unrelated question becomes a General enquiry on the contact timeline and in the team Inbox. It does not create or attach to a job.
- General enquiries waiting on Terra turn amber after 2 working hours and red at end of day. A one-click action can later create a lead/job from the conversation.

13.2 Zoho Mail two-way sync

Event	Required Terra Ops behaviour
Sent from Terra Ops	Send through the connected Zoho mailbox, appear in Zoho Sent, and log immediately to the conversation
Sent directly from Zoho	Sync the Sent folder, de-duplicate by external message ID, then attach to the correct conversation
Incoming email	Receive near real time by webhook/sync, store thread metadata and place in the team Inbox
Existing linked thread	Keep the established contact/job links unless an authorised admin changes them
One open job	Suggest the job; show the link clearly and allow correction
Several possible jobs	Require admin selection; do not guess
Known contact, no job	Attach to contact conversation only
Unknown sender	Place in Unmatched Emails; suggest match/new lead but do not silently create records

- Use provider message IDs and reply/thread headers for de-duplication and threading. A Terra job reference is an additional deterministic match, not the only method.
- Attachments remain on the conversation until an admin files them into a relevant job-file bucket.
- Mailbox permissions are separate: general customer mail may be shared, billing is owner/accounts, and Damien's private mail is not exposed to every admin.

13.3 Marketing automation

- Segment by client type, product interest, location, source, quote/job history, recency and other consented fields.
- Initial automations: lead confirmation, visit reminder, quote follow-up, deposit reminder, pre-install instructions, product-specific aftercare, review request, annual follow-up and lost-lead reactivation.
- Transactional messages and marketing messages are separate. Unsubscribe suppresses marketing without blocking essential job communication where lawful.
- Pause marketing during an unresolved complaint, open defect, payment dispute or no-contact request.
- Enforce consent source, suppression reason, frequency limits, test sends, approval state and full send audit.

14. Voice agent rules and security

The voice agent is built after jobs, CRM, permissions and audit are reliable. Caller ID is a matching hint, not proof of identity.

14.1 Call matching

- Normalise the caller number, find the contact, then show/open recent and active jobs.
- If multiple jobs are possible, ask which address or job the caller means. Append to the existing job only when confirmed.
- If the customer asks a general question, create or continue a contact conversation without creating a job.
- Unknown callers provide name, phone, email, address and reason. Run duplicate checks before creating a draft lead.
- Store recording where permitted, transcript, summary, outcome, linked contact/job/conversation and action items.

14.2 Allowed and prohibited actions

Agent may	Agent must not do without authorised human action
Find a contact/job; create a draft lead; add a note; log complaint/defect	Invent or approve a customer price, discount or variation
Provide a confirmed appointment; offer allowed slots; send confirmation	Promise an installation date outside scheduling rules
Collect details; answer approved FAQs; transfer or request callback	Cancel jobs, issue refunds, change payments or expose sensitive finance data

- For sensitive information, verify an additional fact such as postcode or send a one-time SMS link.
- Low confidence, angry callers, safety issues, legal threats, payment disputes and urgent damage escalate to a human.
- Every agent action records tool input, result, confidence, rule applied and whether a human approved it.

15. Admin permission profiles

Keep the fixed top-level admin and installer roles, but add permission profiles inside admin. Profiles are editable sets of server-enforced capabilities.

Profile	Default access
Owner	Full system; master pricing/rates; margin; low-margin approvals; integrations; audit; exports
Office	Contacts, jobs, schedule, dispatch and shared communications; no private rates unless granted
Estimator	Visits, measures and job quotes; no master-rate edits; no low-margin approval by default
Marketing	Segments, templates, campaigns and reports; no supplier costs, installer rates or job margin
Accounts	Invoices, payments, supplier/accounting records and approved financial reports
Installer	Own offers, assigned tasks, permitted contacts/files and own pay only

Granular permissions

- Separate permissions include: view margin, view supplier cost, view/edit installer rates, change master price, override a job price, approve low margin, send quotes, send campaigns, issue/refund payments, export data, manage users, manage integrations and view global audit.
- Permission changes require owner-level authority, take effect immediately, revoke active sessions where appropriate and create immutable audit events.
- Approval permissions must name the approver; a user cannot approve their own restricted change unless their profile explicitly allows self-approval.

15.1 Data export and deletion

- Bulk export, contact deletion/anonymisation and retention controls are permissioned separately from ordinary viewing.
- Historical job, quote, payment and safety evidence is archived according to retention policy rather than silently hard-deleted.

16. Integration failure and audit system

Every external action is treated as a tracked delivery, not assumed successful. A failed email, SMS, payment, accounting push, file upload or webhook must be visible and recoverable.

16.1 Integration event model

Field	Required behaviour
Provider and operation	Zoho send/sync, SMS, voice, Stripe, Xero, storage, website webhook or other named action
Internal/external IDs	Link the Terra record and provider request/message/payment ID for de-duplication
State	Queued, processing, succeeded, retrying, failed, cancelled or needs human action
Attempts	Attempt count, last attempt, next retry, safe error summary and technical detail
Ownership	Assigned staff member, severity, due time and linked Needs Attention flag
Recovery	Idempotent retry where safe; manual resolve/relink/cancel with required reason

- Automatic retry uses capped backoff and never creates duplicate emails, payments, invoices, leads or files.
- After retry limits, create a red Needs Attention item. Do not leave failures buried in logs.
- Do not show provider secrets or sensitive payloads in the UI, notifications or ordinary logs.

16.2 Immutable audit

- Audit actor, time, action, entity, record ID, previous value, new value, reason, source surface and request/correlation ID.
- High-risk events include login/security changes, permissions, prices/rates, margin approvals, quote/invoice changes, refunds, exports, deletions, integration configuration and AI/voice actions.
- Job activity remains a readable operational timeline; the global audit view is the immutable security and change record.
- Owners can filter/export audit data, but ordinary users cannot edit or delete it.

17. Training, migration and controlled go-live

17.1 In-product training

- Training Mode uses clearly marked fake customers, jobs, prices and communications; it cannot send real messages, charge cards or sync accounting.
- Role-based walkthroughs include a 'What do I do next?' action, searchable help, tooltips and 30-60 second embedded videos.
- Preview and confirmation are mandatory for bulk sends, price changes, imports, refunds and other high-impact actions. Provide undo or version restore where safe.
- Installer onboarding should take about 10-15 minutes: sign in, accept, view, start, photos, variation, complete and check own pay.
- If a normal installer workflow needs lengthy training, simplify the product rather than producing a longer manual.

17.2 Guided setup

- Configure staff/installers and permissions, skills/rates, price book, products/suppliers, templates/terms, lifecycle stages/timers, integrations, data import and one complete test job.
- Do not allow live quoting until minimum margin, GST basis, active prices, installer fallback costs and quote terms are confirmed.

17.3 Migration

- Import contacts/companies, de-duplicate identities, preserve marketing consent/suppression, bring over open jobs and required notes/photos/documents, then select the useful completed history.
- Record source IDs and migration batch so every imported record can be traced and a batch can be safely reviewed.
- Decide which legacy information stays read-only rather than trying to recreate every historical workflow.

17.4 Rollout

Gate	Scope	Exit condition
1	10 historical jobs in Training Mode	Workflows and totals reproduce expected outcomes
2	5 low-risk live jobs run by Damien	No critical data, permission or integration defects
3	One reliable installer	Crew workflow succeeds with poor signal and minimal help
4	All installers and office users	Support issues stable; alerts and ownership working
5	All new enquiries and live integrations	30 stable days before retiring the replaced system

- Avoid long-term dual entry. During transition, declare which system is authoritative for each record type.
- Test backup restore, session/device revocation, offline sync conflicts, duplicate messages, failed payments, duplicate contacts and AI mistakes before broad rollout.

18. Runnable delivery plan and owner input gates

Start Runnable now - do not wait for every price

Runnable can build the platform foundations, lifecycle engine, permissions, audit, CRM/email structure and editable price-book importer now. Prices and products must remain configuration data. The complete lists are required before the estimating and quoting module is accepted for live use, not before development begins.

18.1 Delivery order

Phase	Build now	Owner input needed before acceptance
1. Foundations	Three surfaces, route permissions, audit events, integration event queue	Admin profile assignments and final mobile colour decision
2. Operations	Job lifecycle, Needs Attention, site-visit timer, pending reasons	Business hours, alert owners and timer adjustments
3. CRM/email	Conversations, Inbox, contact timeline, Zoho sync and matching	Mailboxes, visibility, signatures, categories and templates
4. Pricing data	Price-book versions, product/supplier catalogue, bulk importer and draft validation	Full sell prices, installer rates, products, pack sizes, fees and GST basis
5. Estimating	Work packages, areas, calculations, trim optimiser and margin approvals	Minimum margin, waste/kerf rules and quote wording
6. Documents/market ing	Quote documents, campaigns, automations, consent and reporting	Brand samples, terms, campaign copy and approval rules
7. Voice/integrations	Voice agent, Stripe/Xero and recovery workflows	Phone strategy, verified FAQs, financial rules and credentials
8. Go-live	Training Mode, migration tooling, guided setup and rollout controls	Test users, legacy exports and sign-off results

18.2 Price-list handoff

- Damien may provide the lists in this chat in batches. Convert each batch into structured rows with category, item, aliases, unit, sell price, installer cost/rates, supplier cost, GST basis, effective date and notes.
- Products additionally need supplier, brand, SKU, category, technical attributes, purchase unit, pack/box size, coverage, waste factor, current cost and fee rule.
- Import all new rows as Draft. Show validation errors and blanks. Damien reviews and activates them; the AI must not silently repair uncertain product names, units or prices.
- Use the draft seed data in Section 10 for development and test fixtures only. It is not approval to activate live customer or installer pricing.

18.3 Definition of done

- A phase is not complete merely because screens exist. It needs server-side permission tests, calculation tests, failure/retry tests, responsive checks, audit evidence and owner acceptance on realistic jobs.
- Runnable ships each phase behind a feature flag or controlled rollout, with lint/build/tests passing and a rollback or disable path.
- Any proposed change that conflicts with locked decisions in Sections 1-4 or the security rules in Section 10 must stop for owner approval.